

PRANAY BABU TOTAKURA

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Professional Summary

Python Full-Stack Software Engineer with more than two years of expertise in building scalable web applications and automation tools using Flask, REST APIs, and CI/CD. Experienced in deploying microservices on AWS and integrating AI models to drive intelligent workflows. Strong foundation in system design, test-driven development, and backend engineering with a focus on performance, maintainability, and clarity. Passionate about solving real-world problems through clean code, scalable architecture, and AI-driven solutions.

Technical Skills

- Languages & Frameworks: Python, JavaScript, Java, C#, Spring/Spring Boot, TypeScript, Flask, Node.js, React, Angular, HTML5, CSS3
- Cloud & DevOps: AWS Cloud (EC2, S3), Docker, Jenkins, Git, GitHub Actions, Maven, JMeter, Artifactory, GitLab CI/CD, Terraform
- Backend & APIs: RESTful APIs, Microservices, Security (Authentication/Authorization), System Design and Development
- AI/ML: OpenAI GPT-4, LangChain, Scikit-learn, Pandas, Prompt Engineering, Streamlit.
- Testing & QA: JUnit, Pytest, Postman, Selenium, Automated Testing, Integration Testing
- Frontend & UI: React, HTML5, CSS3, Bootstrap, Responsive Design
- Databases & Analytics: PostgreSQL, MongoDB, SQL, Adobe Analytics, Google Tag Manager
- Other: Content Management Systems (CMS), Agile, Scrum, Jira.

Experience

Accenture Solutions Private Ltd., Hyderabad, India

July 2022 – July 2023

Software Engineer – Full-Stack & DevOps

- Developed scalable backend RESTful APIs using Python and C#, accelerating feature delivery by 25%.
- Designed and implemented CI/CD pipelines using Jenkins and GitHub Actions to automate testing and deployment workflows for web applications.
- Deployed containerized microservices on Azure using Docker, improving application portability and managing infrastructure-as-code.
- Built automation tools and engineering dashboards in React and CSS3, reducing manual workload by 20%.
- Wrote comprehensive unit and integration tests using frameworks conceptually similar to JUnit, achieving 85% test coverage.
- Implemented security best practices for web applications, including input validation and authentication/authorization mechanisms.
- Collaborated in an Agile team using Git for version control, participating in code reviews to maintain high standards of code quality.
- Troubleshoot and optimized legacy code, improving website performance and reducing bug rates by 30%.

Projects

Real-Time Developer Portfolio Analytics Dashboard

March 2025 – April 2025

React, Node.js, MongoDB, Chart.js, Heroku

- Built a real-time analytics dashboard using React, CSS3, and Chart.js to track and visualize user engagement metrics, demonstrating expertise in website performance analysis.
- Engineered backend aggregation logic in Python (Flask) and crafted RESTful APIs to serve data to the frontend.
- Containerized with Docker and deployed using Heroku, integrating GitHub Actions for a complete DevOps CI/CD workflow.
- Visualized key metrics with dynamic graphs, enabling data-driven improvements that boosted recruiter engagement by 60%.
- **GitHub:** [ptotakura2023/portfolio-analytics-dashboard](https://github.com/ptotakura2023/portfolio-analytics-dashboard).

StudyPal: GenAI-Powered Study Assistant Bot

October 2024 – December 2024

Python, Discord.py, OpenAI GPT-4, LangChain, MongoDB

- Developed a GPT-powered Discord bot that resolved 1,200+ academic queries with >90% accuracy.
- Improved response precision by 30% through memory-aware routing and dynamic context injection.
- Modularized agent architecture to enable easy plug-and-play of new capabilities within minutes.
- Delivered explainable AI workflows that boosted student productivity and tutoring engagement by 35%.
- **GitHub:** [ptotakura2023/COT_6930_GenAI_project](https://github.com/ptotakura2023/COT_6930_GenAI_project).

Distributed File Ingestion System

March 2024 – June 2024

Python, Flask, AWS S3, Docker, Prometheus, GitHub Actions, Render

- Built a scalable backend system using Python to handle concurrent file uploads to AWS Cloud (S3) with real-time tracking.
- Containerized the application with Docker, reducing setup time by 70%, and deployed with a full DevOps CI/CD pipeline.
- Published versioned Docker images to a private Artifactory registry, ensuring production readiness.
- Implemented robust monitoring, improving system reliability and observability by 80% for seamless upload tracking.
- **GitHub:** [ptotakura2023/distributed-ingestion-system](https://github.com/ptotakura2023/distributed-ingestion-system)

AI Agent Content Optimizer for CMS

September 2023 – November 2023

TypeScript, Node.js, OpenAI API

- Built an AI agent to analyze and enrich content for a Content Management System (CMS), a core competency for this role.
- Deployed REST endpoints using JavaScript (Node.js) to allow frontend teams to query AI-enriched CMS entries on demand.
- Designed and documented the RESTful API using OpenAPI specifications, ensuring clear and consistent usage for development teams.
- Enabled a 40% reduction in manual editorial work, demonstrating experience with optimizing CMS workflows.

Education

Florida Atlantic University, Boca Raton, Florida

August 2023 – May 2025

Master's degree (MS) in Computer Science

Institute of Aeronautical Engineering (IARE), Hyderabad, India

August 2018 – June 2022

Bachelor of Technology (BTech) in Electronics and Communication Engineering

CERTIFICATIONS & COMPETITIONS

Participated in HackerRank coding events such as Python programming and problem-solving.

- Python: [Certificate Link](#)
- Problem solving: [Certificate Link](#)
- Data Structures & Algorithms – HackerRank/LeetCode (150+ problems solved)